



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4007
Design Thinking for A.I. Solutions
Session 5

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Session 5



- Finalizing Prototypes
- Prototype Testing

Finalizing Prototypes



Time: 90 min

- Finalize the Low Fidelity Prototypes in your groups
- Prepare a document (word or powerpoint) as per **A5 Paper Prototype Template** for the usability test (doable in 10 min) with target users
- **MINIMUM REQUIREMENTS** Your prototype should support at least **2 key tasks** and include at least **5 screens or screen states**

Finalizing Prototypes



Assignment 5

Readings

In preparation for this assignment review these:

- <https://www.smashingmagazine.com/2014/10/the-skeetops-guide-to-low-fidelity-prototyping/>
- <https://www.ibm.com/design/thinking/page/toolkit/activity/paper-prototypes>
- <https://video.ibm.com/replayed/108328275> Paper prototype demo – one user Watson Example
- <https://www.youtube.com/watch?v=85muhAxy8os> Different paper prototypes ideas dog bones

Having developed your ideas, prioritized them with respect to the available data, planned the to-be scenario, and hills, you are now ready to start detailing your solution that your target audience will use. You are now ready to create a low-fidelity prototype of your user interface in order to test your early ideas with users as quickly as possible. The goal of low fidelity prototypes is:
Minimal production effort for Maximal user feedback.

In this assignment you will **define the key tasks** that your solution needs to support and **develop the low-fidelity prototype to support those tasks**. During Class 5 you will use your defined tasks to run usability tests of your low-fidelity prototype.

1. Key Tasks

PART 1 – DEVELOP TASKS for TESTING YOUR LOW FIDELITY PROTOTYPE

The video <https://www.youtube.com/watch?v=B7M0VXdoxM> demonstrates a paper prototype of a new Pinterest Mobile App Redesign. In this video you see the TASKS which accompany the prototype. These tasks are used during the prototype testing. The same tasks are also used for defining the scope of the prototype. The low-fidelity prototype will support the key tasks

The scope of your paper prototype is NOT the full functionality of your system. It instead needs to demonstrate some **key aspects of your solution** including the **key tasks** which your target audience will need to perform with your prototype.

For example: If you were developing a solution for the Restaurant Managers. You would NOT prototype the registration nor the login process. Those are pretty standard and do not need to be tested. You would instead include the sections of the solution that present the user important information regarding the key parameters of interest (COVID statistics) that they will use for their decision making process.

PART 2 – CREATE a LOW FIDELITY PROTOTYPE

Once you have defined your solution's key tasks, you are ready to make a low-fidelity prototype.

Low-fidelity prototypes are most easily done on paper, as demonstrated in the above videos. Since we cannot meet in person to test our paper prototypes. We can instead create low-fidelity prototypes and show them digitally to our users and get their reactions. We can still draw the prototype images on paper and then scan them for testing. Alternatively, we can use a sketching tool to produce quick images of the solution to show our users.

Your individual prototype for Assignment 5 **needs to be a low-fidelity prototype!** You can choose your preferred method of creating your prototype. You do NOT need to use prototyping tools for this assignment. Quick clear visualization of your ideas is the goal here. Paper drawings, or digital sketches, that clearly **show your solution** and allow your users to try and interact with it are the requirements.

2. Low-Fi Prototype

3. Test Script

PART 3 – CREATE a SCRIPT for your PROTOTYPE TESTING

In Class 5 we will conduct the usability testing of your Paper Prototype. You will need a script for doing the usability test. Please see Template provided for you to prepare your assignment

A5 Paper Prototype Template in both PPTX and PDF formats.

MINIMUM REQUIREMENTS Your prototype should support at least **3 key tasks** and include at least **5 screens or screen states**.

Finalizing Prototypes



TEMPLATE

Assignment A5 Low-Fidelity Paper Prototype



Examples: Project BearIt, CSC318 2019F

SAMPLE ONLY

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Finalizing Prototypes



A5.1 Key Tasks

1. Create a recurring event
2. View saved activities
3. Change event
4. Delete event

SAMPLE ONLY

Finalizing Prototypes



A5.2 Usability Testing Script **SAMPLE ONLY**

Scenario: You are a 3rd year computer science student. You heard from your friends that there is an app which can help you be more motivated and engaged in your long term goals. You decide to try this app.

Note how the Scenario sets up the participant into their role using a high level description and a realistic situation.

Can you show me how you would do the following tasks using this app:

Note the open ended instruction without too much specific guidance.

1. You want to start working out regularly at the Athletic Centre on Wednesdays, from 8-9pm, starting on October 16. How can you set that up using this prototype?

1. Note how carefully the words are chosen to not bias the participant regarding the interface elements.

2. How can you review the activity that you just added?

2. Note reference to recent activity without using specific UI vocab. Similarly all tasks avoid using any words that appear in the design.

3. You just remembered that you have a midterm coming up so on October 16th you will need to prepare for the midterm instead of working out. How can you update that in your app?

4. You are happy, because your midterm was canceled! How can you remove the preparation activity from your app.

Note the length of the entire test is about 7 minutes. So 4 simple tasks will suffice.

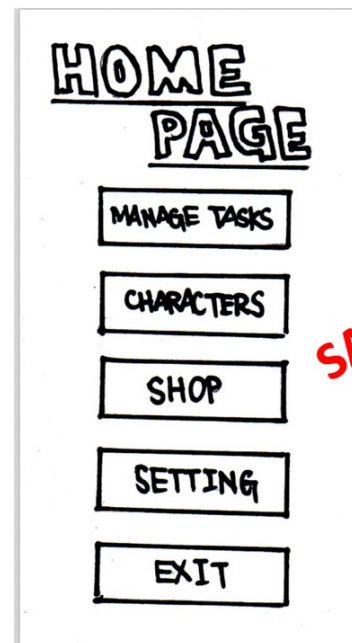
Please think aloud as you interact with the paper prototype.

Note participant may not know to think aloud, nor do this naturally. You may need to remind and probe as they use your paper prototype.

Finalizing Prototypes

A5.3 Low-Fidelity Paper Prototype

Task 1. You want to start exercising regularly at the Athletic Centre on Wednesdays, from 8-9pm, starting on October 16. How can you set that up using this prototype?



SAMPLE ONLY

Note this column of text does not show during the Usability Test Slide Show; but is used for your note taking & record keeping.

Note the task instructions are repeated here to reduce participant's memory load.

EXPECTED USER ACTION

- Click [MANAGE TASKS]

OBSERVED ACTION

USER QUOTES

DESIGN IMPLICATIONS for A5b Results

Finalizing Prototypes

Task 1. You want to start exercising regularly at the Athletic Centre on Wednesdays, from 8-9pm, starting on October 16. How can you set that up using this prototype?

MANAGE TASKS

October, 2019

SU	MO	TU	WE	TH	FR	SA
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Oct, 16 (WED)
No Task

SAMPLE ONLY

Note this column of text does not show during the Usability Test Slide Show; but is used for your note taking & record keeping.

EXPECTED USER ACTION

- Click [October 16]
- Click [ADD +]

OBSERVED ACTION

USER QUOTES

DESIGN IMPLICATIONS for A5b Results

Finalizing Prototypes



Task 1. You want to start exercising regularly at the Athletic Centre on Wednesdays, from 8-9pm, starting on October 16. How can you set that up using this prototype?

NEW TASK Oct 16
WED

From 00:00 To 00:00

Location

Type of Task

☐ Lecture ☐ Study

☐ Exercise

Others

☐ Alarm ☐ Weekly

⬅️ ➡️

NEXT

Note this column of text does not show during the Usability Test Slide Show; but is used for your note taking & record keeping.

EXPECTED USER ACTION

- Type FROM [20:00]
- Type TO [21:00]
- Type LOCATION [Athletic Centre]
- Check [EXERCISE]
- Check [Weekly]
- Check [Alarm (optional)]
- Click [CHECK MARK]

NOTE it's good practice to ask users "What do you expect to happen next?"

OBSERVED ACTION

USER QUOTES

DESIGN IMPLICATIONS for A5b Results

[NEXT] button is an **Action Button** you can program in Powerpoint for navigation.

Prototype Testing



Time: 45 min

- Use your defined tasks to run usability test of your low-fidelity prototype
- Conduct the usability testing of your Paper Prototype. You will need a script for doing the usability test (see **A5 Paper Prototype Template**)
- Make a note of the user feedback on MURAL